From Scalar Charge to Local Force

Testing spherical chameleon thin-shell models

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Code, numerical data, tests, and reports

1. Charge accuracy and force accuracy are separate questions

Asymptotic source description

$$u = 1 - sQ\frac{e^{-\mu(x-1)}}{x}$$

A single charge summarizes the distant field.

- Does a thin-shell model predict Q?
- Which correction improves that prediction?

Classical-mechanics observable

$$a_5 = -\frac{\beta}{M_{\rm Pl}}\nabla\phi$$

A nearby probe responds to the local gradient.

- How far away is the tail formula accurate?
- Does the approximation change a threshold decision?

The numerical force reference is the **direct nonlinear gradient**.

2. Spherical model and the thin-shell parameter

$$V=\frac{\Lambda^5}{\phi},\quad x=\frac{r}{R},\quad u=\frac{\phi}{\phi_\infty}$$

$n=1$, weak conformal-coupling limit

$$u''+\frac{2}{x}u'=\frac{\mu^2}{2}(\tilde{\rho}-u^{-2})$$
$$u'(0)=0,\qquad u(\infty)=1$$

$$\delta=\frac{\rho_c}{\rho_\infty},\quad \mu=m_\infty R,\quad m_c R=\mu\delta^{3/4}$$

$$\epsilon_{\rm th}=\frac{2(1-\delta^{-1/2})}{\mu^2\delta}$$

This potential-defined parameter is not an exact geometric thickness.

Finite interior-mass correction

$$\epsilon_{\rm th}\simeq\frac{\Delta R}{R}+\frac{1}{m_c R}$$

Relevant large-mass asymptotics: Tamaki–Tsujikawa (2008), Eq. (48).

The full exponential coupling has $D=\phi_\infty/M_c$; the scan explicitly takes $D\to 0$.

3. Finite-range models and a matched ablation control

$$Q_{\mathrm{LR}} = \min(3\epsilon_{\mathrm{th}}, 1), \qquad Q_{\mathrm{surf}} = Q_{\mathrm{LR}}, \quad Q_{\mathrm{ctr}} = e^{-\mu} Q_{\mathrm{LR}}$$
$$q = \sqrt{\max(1 - 2\epsilon_{\mathrm{th}}, 0)}, \qquad Q_{\mathrm{MS}} = 1 - q^3$$

Point factor + finite volume

$$Q_{\mathrm{PV}} = e^{-\mu}(1 - q^3)$$

Same assumed core and active volume.

Additional spherical radial weight

$$Q_{\mathrm{AS}} = \frac{3e^{-\mu}}{\mu^3}[g(\mu) - g(\mu q)]$$
$$g(z) = z\cosh z - \sinh z$$

All coefficients use one surface-tail convention. Changing reference alone does not change physics; the first two formulas are different amplitude hypotheses. The core radius is modeled, not measured.

4. Nonlinear BVP and the direct force reference

- Exact density jump: separate inner and outer domains
- Regular center, matched surface, exterior Robin condition
- Charge-scaled linear/log field variables
- Adaptive fourth-order SciPy collocation

Numerical experiment

725 BVPs, $x_{\max} = 160$
$10^2 \le \delta \le 10^5$, $10^{-3} \le \mu \le 0.3$
71 direct probe radii from $1.02R$ to $20R$

Local nonlinear force

$$\alpha_{\mathrm{direct}}(x) = \frac{2\beta^2 Q_t}{s} x^2 u'_{\mathrm{num}}(x)$$

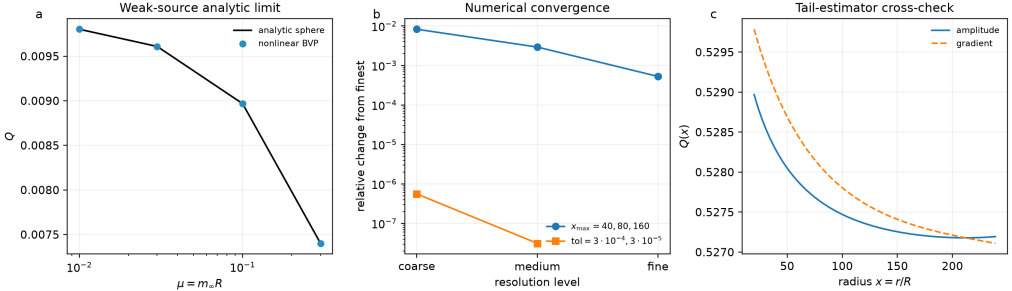
$$s = \frac{\mu^2 \delta}{6}$$

The tail model is evaluated separately:

$$\alpha_{\mathrm{tail}} = 2\beta^2 Q_t Q(1 + \mu x) e^{-\mu(x-1)}.$$

Reference: the body's total-mass Newtonian acceleration.

5. Analytic and refined-resolution checks

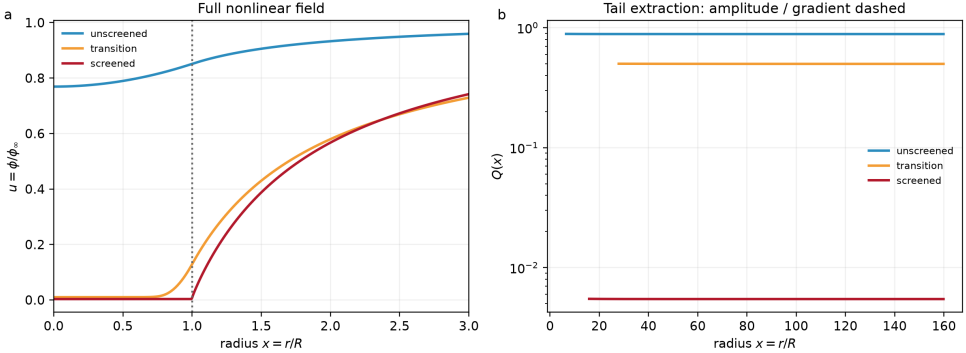


Refined audit: 16 points, including local-force threshold flips.

Charge change: at most 0.056%. Local-force scan/refined difference: at most 0.0026%.

Direct-force decision changes in the audit: **0**.

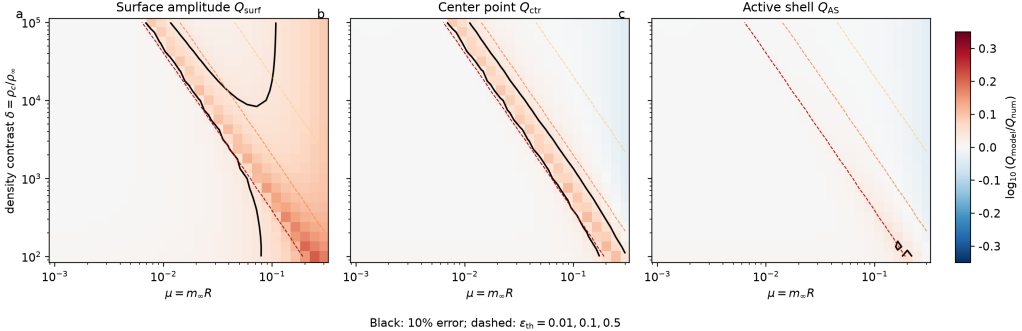
6. The field and its asymptotic charge



The far-tail amplitude and gradient estimators agree.

That agreement checks charge extraction, not near-source linearity.

7. Charge accuracy on the parameter grid



Black: 10% error; dashed: $\varepsilon_{\rm th} = 0.01, 0.1, 0.5$

10% charge coverage: surface 68.8%, center 91.3%, active shell 99.7%.

These are fractions of the chosen grid, not probabilities for a theory.

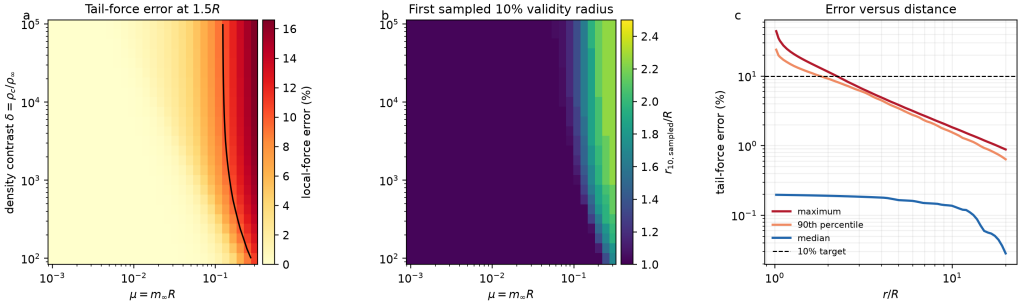
8. The simpler control reaches the same charge coverage

Model	Median error	Maximum	Within 10%
Surface-amplitude model	2.78%	62.6%	499/725
Center-point model	0.566%	30.1%	662/725
Finite shell volume	2.18%	37.6%	562/725
Point + finite volume	0.378%	12.7%	723/725
Active-shell integral	0.391%	13.1%	723/725

$$Q_{\mathrm{PV}} = e^{-\mu} Q_{\mathrm{MS}} \qquad \text{versus} \qquad Q_{\mathrm{AS}}$$

- Additional radial weighting changes the control by at most 1.51% here.
- Both reach 99.7% coverage; radial weighting is not established as necessary.

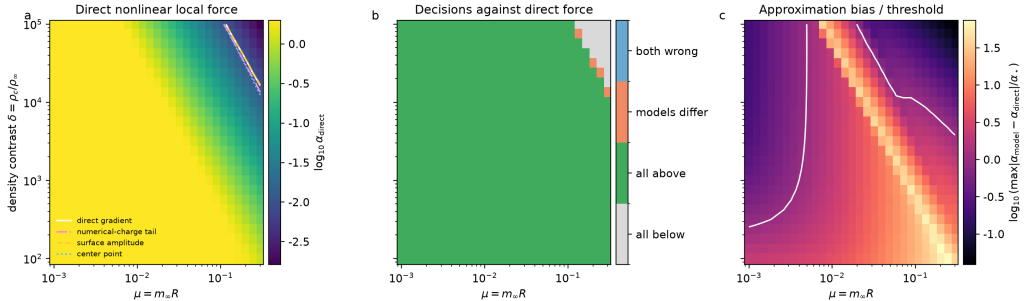
9. Even the numerical charge can mispredict local force



Worst tail-force error: **16.6% at** $1.5R$, 3.85% at $5R$, 1.84% at $10R$.

The first-validity radius tests every later sampled radius through $20R$.

10. Threshold map corrected to the direct force



$\beta = Q_t = 1$, $r = 1.5R$, illustrative threshold $\alpha_\star = 10^{-2}$.

Numerical-charge tail: 2 wrong decisions. Surface / center models: 1 / 3.

This is a local force-ratio map; apparatus scale and noise are not included.

11. Error cancellation changes the observable ranking

$$\frac{\alpha_{\mathrm{tail}}(Q_j, x)}{\alpha_{\mathrm{direct}}(x)} = \frac{Q_j}{Q_{\mathrm{num}}} \frac{\alpha_{\mathrm{tail}}(Q_{\mathrm{num}}, x)}{\alpha_{\mathrm{direct}}(x)}$$

Two errors multiply

A charge overestimate can partly cancel a near-field tail underestimate. Charge accuracy alone does not rank local-force predictions.

Active-shell 10% coverage:
charge 99.7%, force at $1.5R$ 85.2%.

- Test several force-ratio thresholds; no crossing is a valid result.
- Distance validity is conditional on this $n = 1$, $D \to 0$ spherical grid.
- Real detectability also needs geometry, modulation, probe response, and noise.

12. What this study establishes

1. Specified charge models can be benchmarked reproducibly against a nonlinear BVP.
2. The ablation limits the causal claim for radial weighting on this grid.
3. A local gradient reference reveals a distance-dependent propagation error.
4. The corrected force changes 2 decisions at $\alpha_\star = 10^{-2}$.

Next research question

How does this charge-to-force validity change for $\mu \gtrsim 1$, chamber walls, or different source shapes?

Code, data, tests, and both language editions on GitHub

Selected references

[1] J. Khoury and A. Weltman, *Phys. Rev. Lett.* 93, 171104 (2004); *Phys. Rev. D* 69, 044026 (2004).

[2] T. Tamaki and S. Tsujikawa, *Phys. Rev. D* 78, 084028 (2008).

[3] C. Burrage and J. Sakstein, *Living Rev. Relativ.* 21, 1 (2018).

[4] B. Elder et al., *Phys. Rev. D* 94, 044051 (2016).

[5] S. Schlögel et al., *Phys. Rev. D* 93, 104036 (2016).

[6] D. Sabulsky et al., *Phys. Rev. Lett.* 123, 061102 (2019).

[7] P. Yin et al., *Nature Phys.* 18, 1181–1185 (2022).

Full derivations, limitations, and reproduction instructions are in the report.